i-CAPE

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Innovative Science Olympiad

2026-2027

Grade 5

Total Questions: 30 | LEVEL – II | Set - B | Time: 1.5 hr. | Total Marks: 60

SECTION - 1	• This section contains **Twelve (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer. • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.
SECTION - 2	• This section contains **Six (06)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s). • For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given. • There are **NO NEGATIVE** marks for wrong answers.
SECTION - 3	• This section contains NINE (09) questions of PASSAGE TYPE OR COMPREHENSION TYPE. • Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer. • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.
SECTION - 4	• This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS.** • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers

Section - 1

1. While helping in the kitchen, a student accidentally touches a hot vessel and gets a minor burn on his finger. Which of the following is the most appropriate first-aid step?

A) Rubbing hot oil or butter on the burn to soothe the pain.

B) Placing the burned area under cool running water for at least 10 minutes.

C) Pricking any blisters that form with a needle to let the fluid out.

D) Wrapping the finger tightly with a thick, warm woollen cloth.

2. A chef mixes the following ingredients to make three different items:

- Item A: Chocolate syrup mixed with milk
- Item B: Nuts, oats, and honey
- Item C: Oil and vinegar in a salad dressing

Match them with homogeneous, heterogeneous, and immiscible mixtures:

A) Item A – Homogeneous, Item B – Heterogeneous, Item C – Immiscible

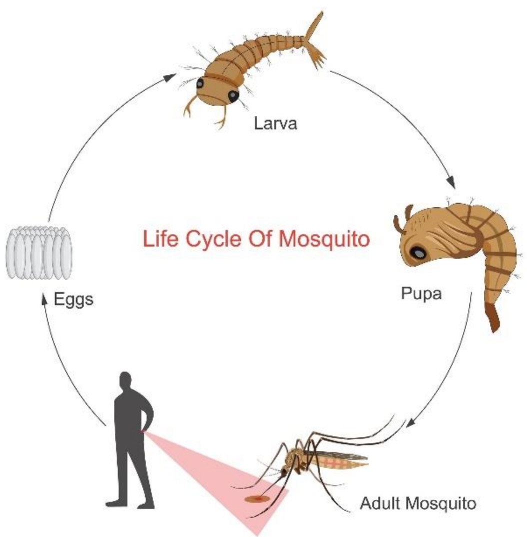
B) Item A – Heterogeneous, Item B –

Homogeneous, Item C – Immiscible
C) Item A – Homogeneous, Item B – Immiscible, Item C – Heterogeneous
D) Item A – Immiscible, Item B – Heterogeneous, Item C – Homogeneous

3. **Why is 'Biomass' considered a renewable resource, while 'Coal' is considered non-renewable, even though coal also comes from ancient plants?**

A) Biomass produces more heat energy than coal per kilogram.
B) Biomass can be replaced in a short human timeframe, whereas coal takes millions of years to form.
C) Biomass does not release carbon dioxide when burned.
D) Coal is found deep underground, which makes it non-renewable.

4. **Observe the life cycle above. A village removes stagnant water from old tyres and flowerpots during the rainy season. How does this help prevent diseases like dengue and malaria?**



A) It kills the germs present in water directly
B) It improves soil quality
C) It cleans the air around the village
D) It stops mosquitoes from breeding in water

5. **The diagram shows two types of leaves-one single blade and another divided into parts. Which pair correctly identifies them?**

A) Simple leaf and compound leaf
B) Single leaf and multiple leaves
C) Lamina and leaflet
D) Stem leaf and branch leaf

6. **If Earth had no tilt, what would MOST likely happen?**

A) No day and night B) No seasons
C) No Moon phases D) No eclipses

7. **A lab mixture contains:**

- Iron filings
- Sand
- Salt

The correct separation methods are:
A) Handpicking → Filtration → Evaporation
B) Magnet → Filtration → Evaporation
C) Decantation → Handpicking → Filtration
D) Filtration → Magnet → Distillation

8. **A geologist finds a rock:**

- Hard and strong
- Has crystals
- Formed from cooling magma

Which rock is it, and where was it formed?

A) Basalt – inside Earth
B) Obsidian – outside Earth
C) Granite – inside Earth
D) Shale – outside Earth

9. **Two animals show similar body structure, yet one survives in water while the other thrives on land under different conditions. Which factor BEST explains this difference in survival?**

A) Variation in food availability alone
B) Adaptation to specific habitats
C) Difference in external appearance
D) Variation in body size

10. **Assertion (A): Comets develop tails only when they are near the Sun.**

Reason (R): The Sun's heat causes ice in comets to vaporize and form a glowing tail.

A) Both A and R are true, and R correctly explains A
B) Both A and R are true, but R does NOT explain A
C) A is true, but R is false
D) A is false, but R is true

11. **A student observes that iron objects left in the open develop rust over time. Which gas in the atmosphere is directly responsible for this process?**

A) Nitrogen B) Oxygen
C) Carbon dioxide D) Helium

12. **While drawing water from a well using a pulley, a person pulls the rope downward. Which statement is correct?**

A) Only push force is used
B) Only pull force is used
C) Both push and pull forces are used indirectly
D) No force is involved

Section - 2

13. **The human body maintains balance and coordination during activities like walking or standing on one leg. Which of the following help in maintaining balance?**

A) Inner ear helps in balance
B) Eyes help in judging position
C) Muscles and joints support movement
D) Teeth control balance

14. **Which of the following are actual effects that a force can produce on an object?**

A) Change the speed of a moving object.
B) Change the direction of a moving object.
C) Change the mass of the object.
D) Change the shape of a stationary object.

15. **Which of the following represents the correct arrangement of substances in increasing order of forces of attraction between their particles?**

A) Oxygen < Water < Sugar

B) Water < Sugar < Oxygen
C) Sugar < Water < Oxygen
D) Oxygen < Sugar < Water

16. During an earthquake, some buildings collapse while others remain safe.

Engineers study the reasons for this difference.
Which factors affect the damage caused by earthquakes?

A) Strength and design of buildings
B) Type of soil and ground stability
C) Colour of buildings
D) Intensity of seismic waves

17. Imagine a swinging pendulum in a giant clock. At the exact moment the pendulum reaches its highest point on the left side and stops for a split second.

A) Its Kinetic Energy is at its maximum value.
B) Its Potential Energy is at its maximum value.
C) Its Kinetic Energy is zero.
D) It possesses only Mechanical Energy in the form of Potential Energy.

18. A student observes ice melting into water and then turning into steam on heating. Which statements are correct?

A) Heat energy increases particle motion
B) Distance between particles increases
C) The substance changes into a new material
D) These are reversible physical changes

Section - 3

COMPREHENSION-1

A farmer uses water for irrigation, and a nearby factory uses water in cooling machines and chemical processes. Both depend heavily on water for their work.

19. Why is water essential for crops in the farmer's field?

A) It cools the soil
B) It is needed for irrigation and germination of seeds
C) It generates electricity
D) It cleans machinery

20. The factory uses water in car radiators to prevent engines from overheating. Which property of water is being utilized here?

A) Water's ability to dissolve salts
B) Water's ability to conduct electricity
C) Water's high heat capacity and cooling property
D) Water's gaseous state

21. If the river supplying water to both the farm and factory becomes heavily polluted, which combination of problems might occur?

A) Health hazards, poor crop growth, and disrupted industrial processes
B) Increased rainfall and better electricity generation
C) Less evaporation from plants and cooling issues in radiators
D) More ice formation and improved seed germination

COMPREHENSION-2

Every simple machine facilitates work by altering the magnitude or direction of an applied effort, though the total work performed remains constant regardless of the mechanical advantage. Friction acts as a dual-natured force, simultaneously resisting motion and providing the necessary grip for machines like screws and wheels to function.

22. If a car is stuck on a patch of "perfect ice" (zero friction) and the wheels are spinning rapidly, what is the state of "Work" and "Force" in this system?

A) Muscular/Engine force is being converted into useful work.
B) Friction is resisting the motion, so the car moves slowly.
C) Force is being applied, but zero work is being done on the car because there is no displacement.
D) The gravitational force has increased, pinning the car to the ice.

23. If a magical lubricant were applied to a standard metal screw and a wooden block, reducing friction to absolute zero, what would be the most likely result when trying to use the screw?

A) The screw would enter the wood with zero effort and stay there forever.
B) The screw would become a more efficient wedge and cut the wood better.
C) The screw would lose its ability to hold the wood together and would slip out instantly under any tension.
D) The screw would turn into a wheel and axle system.

24. If the lever is designed so that the worker only needs to apply 1/4th of the actual weight of the beam as effort, which of the following must be true?

A) The worker does 4 times less work than if he lifted it by hand.
B) The worker must move his end of the lever 4 times the distance that the beam rises.
C) The lever has reduced the gravitational pull on the beam by 75%.
D) The work output has become 4 times greater than the work input.

COMPREHENSION-3

While walking on the road, Rahul suddenly saw a car coming fast and quickly moved aside. His eyes detected the danger, and his brain sent signals to his muscles to act immediately.

25. Which organ helped Rahul detect the danger?

A) Skin B) Eyes
C) Heart D) Lungs

26. How did Rahul move quickly away?

A) Muscles received signals from the brain
B) Blood stopped flowing
C) Lungs pushed the body
D) Bones moved automatically

27. Which system helps in coordination of body actions?

A) Digestive system
B) Excretory system
C) Respiratory system
D) Nervous system

Section - 4

28. Match Column A with Column B

Column A (Real-life Scenario)	Column B (Concept/Action)
(A) A region receives heavy rainfall and has thick green vegetation all year	(P) Improves air quality
(B) Forests releasing oxygen and	(Q) Tropical rainforest

absorbing carbon dioxide	
(C) Forests providing fruits, wood, and medicines	(R) Deforestation
(D) Cutting trees on a large scale for construction	(S) Resource provider

Which of the following options shows the correct matching?

A) A-Q, B-P, C-S, D-R

B) A-R, B-Q, C-P, D-S

C) A-Q, B-S, C-P, D-R

D) A-P, B-R, C-S, D-Q

29. Match Column A with Column B

Column A	Column B
(A) Asthma	(P) Digestive system
(B) Diarrhoea	(Q) Respiratory system
(C) Dengue	(R) Circulatory system
(D) Jaundice	(S) Liver function

A) A-P, B-Q, C-R, D-S

B) A-Q, B-P, C-R, D-S

C) A-Q, B-P, C-S, D-R

D) A-Q, B-P, C-R, D-S

30. Match Column A with Column B

Column A	Column B
(A) Uranium powers a submarine	(P) Natural electrical energy. No pollution.
(B) Pushing a brick wall all day.	(Q) Earth's inner heat. Eco-friendly and safe.
(C) Geothermal plant warms a house.	(R) Uses up body energy. Absolutely no work happens.
(D) Lightning strikes the ground.	(S) Non-renewable energy. Dangerous to extract.

A) A–S, B–R, C–Q, D–P

B) A–Q, B–R, C–S, D–P

C) A–S, B–P, C–Q, D–R

D) A–P, B–R, C–Q, D–S